INDIAN INSTITUTE OF TECHNOLOGY MADRAS
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
Ref. No.: IITM/CSE/ACAD/2026/34 Date: 11 September 2026

NOTICE

Preparatory Academic Requirements for CS2700 – Programming and Data Structures (PDS)

All Second-Year students are hereby informed that, owing to the revised academic schedule and the limited preparation period available prior to the commencement of regular Semester III coursework, students are required to undertake independent preparatory study for CS2700 – Programming and Data Structures (PDS).

The present notice sets out the prescribed preparatory scope, programming requirements, recommended learning resources, coding environment, and GitHub-based academic work applicable specifically to CS2700. Students are advised to commence their preparation immediately and to make effective use of the resources provided by the Institute.

1. COURSE INFORMATION

Particulars	Details
Course Code	CS2700
Course Title	Programming and Data Structures (PDS)
Department	Computer Science and Engineering

CS2700 forms a core programming and algorithmic foundation, progressing from advanced C programming and memory management to abstract data types and fundamental linear and non-linear data structures. The preparatory syllabus is divided into six modules.

2. MODULE 1 – ADVANCED C CONCEPTS & MEMORY MANAGEMENT

Pointers and Pointer Operations

- Pointer arithmetic
- Pointers to pointers
- Function pointers

Dynamic Memory Allocation

- malloc( )
- calloc( )
- realloc( )
- free( )
- Dynamic allocation and deallocation
- Memory leaks
- Dangling pointers

Structures and Memory Organisation

- Structures
- Unions
- Memory alignment
- Memory padding

Abstract Data Types and Modular Programming

- Abstract Data Types (ADTs)
- ADT implementation concepts
- Modular programming
- Separation of functionality into appropriate program components

3. MODULE 2 – LINEAR DATA STRUCTURES

Linked Lists

- Singly linked lists
- Doubly linked lists
- Circular linked lists
- Skip lists
- Node creation, traversal, insertion, deletion and appropriate memory management

Stacks

- Array-based implementation
- Pointer-based implementation
- Stack operations
- Parenthesis balancing
- Infix-to-postfix conversion
- Expression evaluation

Queues

- Linear queues
- Circular queues
- Deques
- Priority queue interfaces

4. MODULE 3 – SORTING AND SEARCHING ANALYSIS

Asymptotic Analysis

- Big-O notation
- Big-Ω notation
- Big-Θ notation

Sorting Algorithms

- Merge Sort
- Quick Sort – deterministic variant
- Quick Sort – randomized variant
- Heap Sort
- Counting Sort
- Radix Sort

Searching

- Binary Search
- Variants of Binary Search
- Order statistics

5. MODULE 4 – TREE STRUCTURES

Binary Trees

- Binary-tree representation
- Pre-order traversal
- In-order traversal
- Post-order traversal
- Level-order traversal

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Binary Search Trees

- Search
- Insertion
- Deletion
- Predecessor
- Successor

Balanced Binary Search Trees

- AVL Trees
- Balance factors
- Rotations
- Red-Black Trees

Heaps

- Min-heaps
- Max-heaps
- Build-heap operations
- Priority queues

6. MODULE 5 – HASHING

Hashing

- Hash functions
- Hash-table fundamentals
- Collision resolution
- Chaining
- Linear probing
- Quadratic probing
- Double hashing
- Load-factor analysis

7. MODULE 6 – GRAPH FUNDAMENTALS

Graph Representation

- Adjacency matrix
- Adjacency list

Graph Traversals

- Breadth-First Search (BFS)
- Depth-First Search (DFS)

Applications

- Cycle detection
- Topological sorting
- Connected components
- Bipartite testing

8. PROGRAMMING LANGUAGE

The primary programming language for the preparatory work prescribed under CS2700 shall be C.

- Write programs
- Compile programs

- Execute programs
- Test programs using appropriate inputs
- Identify programming errors
- Debug programs
- Implement the data structures and algorithms covered in the prescribed syllabus

Particular emphasis shall be placed on pointers, dynamic memory allocation, structures and memory management, as these concepts form an important foundation for subsequent data-structure implementation.

Students may use C++ or other programming languages for additional conceptual understanding. However, students should give primary attention to C-based implementation for the preparation prescribed under this notice.

9. WHERE STUDENTS SHOULD CODE

Students may practise and develop their programs using a suitable local development environment. The Department recommends:

- Visual Studio Code
- GCC/GDB
- Linux or WSL, where available
- Any other standard C-compatible development environment

Students are encouraged to become familiar with compiling and running C programs through the terminal in addition to using an IDE. The ability to understand compilation errors, runtime errors and debugging procedures is considered an important part of the preparation.

10. GITHUB ACCOUNT AND ACADEMIC CODE REPOSITORY

Programming work associated with CS2700 shall be maintained using the GitHub account provided/designated by IIT Madras, wherever the Institute has prescribed GitHub-based academic work.

Students shall maintain their programming exercises and implementations in the repository designated for them. The repository should be maintained in an organised manner. Students are advised to:

- Use meaningful file names
- Organise programs according to topics/modules
- Maintain readable source code
- Include appropriate comments where necessary
- Commit work regularly
- Ensure that submitted programs compile and execute correctly
- Preserve the repository in accordance with the instructions issued by the Department

The exact GitHub repository address, repository structure, submission mechanism, assignment-wise requirements and deadlines shall be communicated separately through the official course channels.

Students should not create or use an alternative repository in place of the Institute-designated repository where a specific submission repository has been assigned.

11. CODING AND ACADEMIC INTEGRITY

Students are expected to develop their programming skills through independent practice.

Students may consult books, official documentation, lectures, tutorials, online courses, programming references and other legitimate educational resources. However, students should not submit copied source code as their own academic work.

Where external code or reference implementations are consulted for learning, students should ensure that their final academic work complies with the applicable Institute requirements concerning originality and academic integrity.

Any assignment-specific rules concerning collaboration, permitted references, or use of external material shall be communicated separately by the concerned instructor.

12. RECOMMENDED FOUNDATIONAL COURSE – CS50x

Students are strongly encouraged to complete CS50x – Introduction to Computer Science, offered by Harvard University, as supplementary preparation for CS2700.

The course may be particularly beneficial for students who require additional strengthening in:

- C programming
- Programming fundamentals
- Problem solving
- Algorithms
- Memory
- Data structures
- Computational thinking

Students who have already completed CS50x are not required to complete the course again.

Such students may retain their completion records and relevant GitHub work for future reference.

If specifically requested by IIT Madras, students who have already completed CS50x may be required to provide their GitHub account details, relevant GitHub repository address, and appropriate course-completion information.

CS50x is supplementary preparation only and does not constitute a replacement for CS2700 or any requirement prescribed by IIT Madras.

13. USE OF THIRD-PARTY RESOURCES

Students may use suitable third-party educational resources for additional learning. Such resources may include books, online lectures, recorded courses, programming tutorials, documentation and practice platforms.

Students should ensure that the material studied corresponds to the CS2700 syllabus and learning objectives communicated by the Institute.

Where IIT Madras has provided a specific learning resource, students should treat the Institute-provided resource as the primary reference.

14. EXPECTED PREPARATORY SEQUENCE

C Programming Fundamentals → Pointers & Pointer Operations → Dynamic Memory Management → Structures & ADTs → Linked Lists → Stacks & Queues → Sorting & Searching → Trees & Heaps → Hashing → Graphs

15. IMPORTANT INSTRUCTION

The purpose of this preparatory requirement is to ensure that students have sufficient familiarity with the fundamental concepts of programming and data structures before regular CS2700 instruction progresses into more advanced material.

Students are not expected to wait until regular classes commence before beginning their preparation.

The Department strongly recommends that students begin with C programming, pointers, structures and dynamic memory management, particularly if they do not currently possess confidence in these areas.

Further instructions concerning CS2700 assignments, course resources, GitHub repositories, submission procedures, assessment and laboratory-related academic requirements, wherever applicable, shall be issued separately through the official departmental/course communication channels.

This notice is restricted to CS2700 – Programming and Data Structures. No preparatory requirements for any other Semester III course are prescribed through this notice.

Students are advised to rely exclusively upon the latest official communication issued by the Institute and the Department.

By Order of the Head of the Department
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